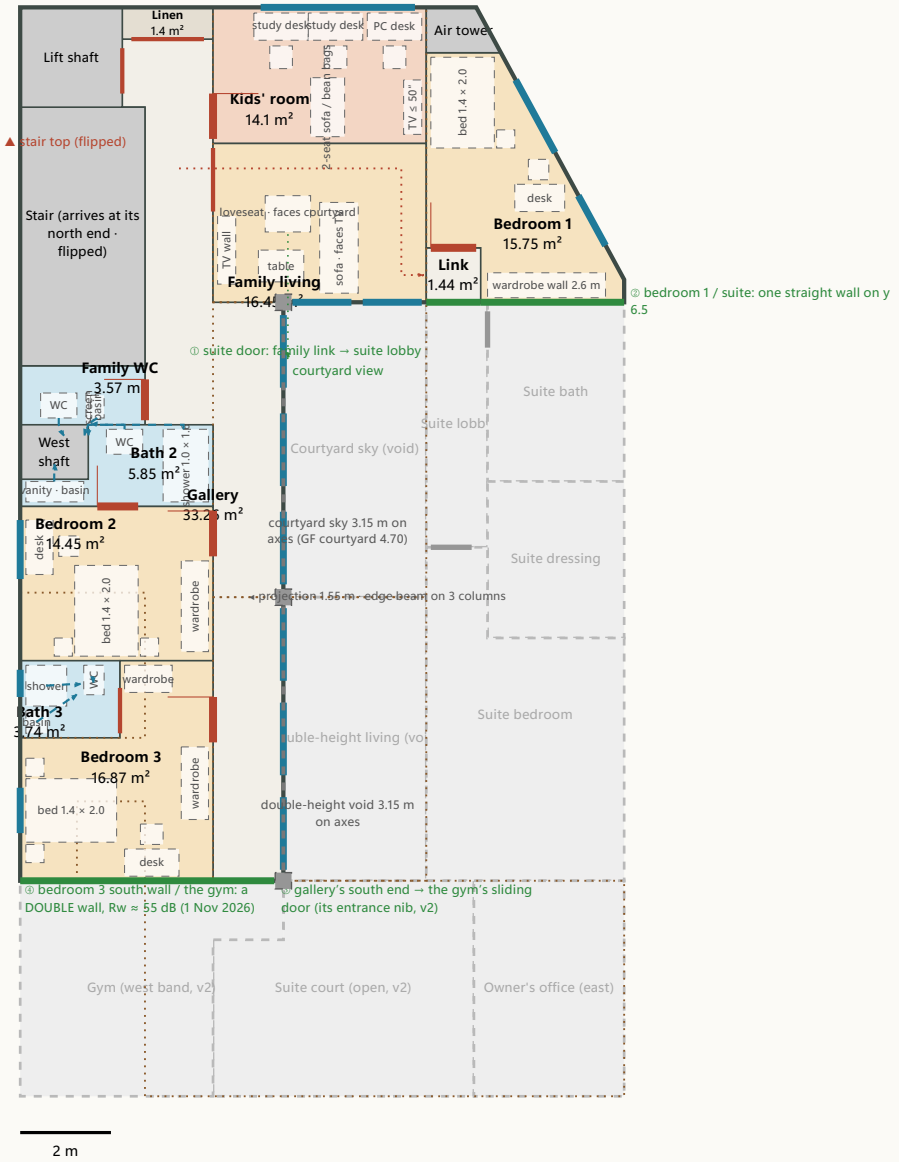
First floor · family part · focus study

Taif / Palm House · first floor V6.3 · local = main CSV coordinates (x east, y south) · wall centre lines · north up

First floor V7.1 · part 1 of 2 · W1 as six arched windows (V7.1): family part · west projection 1.55 m on three columns, the gallery and the family living glazed to the courtyard (W1, W2), west rooms 4.25 m wide · part 2 (office, gym, suite court, master suite): studies/ff_suite.typ



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Gallery (north part + on the projection)	polygon	33.26	29.95
Linen cupboard	2 × 0.7	1.4	–
Family WC (gallery)	2.75 × 1.3	3.57	2.73
Bedroom 2 bath (en-suite)	polygon	5.85	4.63
Bedroom 2	4.25 × 3.4	14.45	12.84
Bedroom 3 bath (en-suite)	2.2 × 1.7	3.74	2.92
Bedroom 3	polygon	16.87	14.97
Kids' room (study + gaming)	4.7 × 3	14.1	12.44
Family living	4.7 × 3.5	16.45	15.16
Family link	1.2 × 1.2	1.44	1.08
Bedroom 1	polygon	15.75	13.83
Fixed: stair, lift, west shaft, air tower		23.8	
Zone		**150.69**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, ≈ 50 mm balustrade on the void edges, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the routes · door swings clear · every room has a door reached from circulation · routes ≥ 1.0 m clear · suite door and gallery end on circulation

Key numbers

- **West projection:** 1.55 m over the courtyard and the double-height void (x 4.25–5.8, y 6.5–19.25) on an **edge beam and 3 columns** at x 5.8 the first-floor west wall stands on the GF west wall. Stair, lift, landing and west shaft rise from the ground floor.
- **Courtyard sky** and **double-height void** 3.15 m on axes (the GF courtyard is 4.70); the ground-floor courtyard and living keep their full floor.
- **West rooms** 4.25 m on axes. **Gallery on the projection:** 1.55 m (½ wall + ½ railing off); wider at the stair top. **Circulation in the zone:** 34.7 m² (landing, gallery, family link); the walk through the living is not counted.
- **Family living:** 16.45 m², on the courtyard edge, no separate route. **Kids' room:** 14.1 m².
- **Bedrooms:** 2 · 14.45 m² · 3 · 16.87 m² + its own bath 3.74 m² · 1 · 15.75 m² (no bath). **Bedroom 2's bath** 5.85 m² and the **family WC** 3.57 m² (3 Nov 2026; the family WC serves bedroom 1, the family living and the kids' room: no shower for them on this floor).
- **Walked to a bathroom door:** bedroom 2 → its own bath ≈ 2.7 m · bedroom 3 → its own bath ≈ 2.7 m · family living → the family WC ≈ 6.4 m · bedroom 1 → the family WC ≈ 14.9 m (no shower on its route: see its study).

Plan: red = doors (thin = opening without a leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below (the full courtyard and living floors) · grey dashed = the edge beam (x 5.6) and the three ground-floor columns · grey = neighbours · dashed boxes = furniture and fixtures.

Family part · the west projection and the rooms

The west projection · 1.55 m over the courtyard and the double-height void

- **Where:** along the west edge of the courtyard and of the double-height living, x 4.25–5.8, from the courtyard's north edge (y 6.5) to the living's south wall (y 19.25). The west rooms take the full band over the GF west band and riwaq; their west wall stands on the GF west wall.
- **Widths:** rooms 4.25 m on axes (≥ 4.0 m clear), gallery 1.55 m (≥ 1.0 m clear).
- **Carried on:** an edge beam along x 5.8 and 3 columns, each in an existing wall or glass line, none free-standing in a room: column 1 in the women's majlis south wall (y 6.5), column 2 a pier in the glass between the courtyard and the living (y 13.0), column 3 in the living's south wall (y 19.25). On the ground floor it reads as a colonnade along the courtyard edge.
- **Why columns and not a cantilever wall:** walls stand directly on walls; the earlier split projection (a west wall on cantilever beams, 0.11 m from the M line) was reverted.
- **Gallery:** closed to the courtyard by the steel window W1 on the upstand of the edge beam (sill +3.82, guarding transom +4.70, head +6.75, four inward casements; V6.4, enclosure option A), the stair arrives at its north end beside the lift door (the flip, option B); every west room door opens from it. Clear 1.405 m (1.235 m at the three column sleeves Ø0.40).

Kids' room · family living · bedroom 1

- **Kids' room** 14.1 m²: two study desks + PC desk under the north window, a gaming corner (TV on the east wall), solid door from the gallery.
- **Family living** 16.45 m²: a view, not a path; the walk to the family link passes behind and beside the sofa.
- **Bedroom 1:** one room, no bath (option A, 2 Nov 2026), its south wall straight on y 6.5. **3 Nov 2026:** its "hotel layout" en-suite (B4) was studied and failed the acceptance criteria (`studies/bedroom1_bath.typ` page 3): until a bath is found for it, bedroom 1 has only the family WC on this floor (no shower).

Family WC · 2.75 × 1.3 m, by the west shaft, its door from the gallery

- **3 Nov 2026 (owner):** the F-b bath is split. The north strip (y 7.9–9.2, ≈ 2.8 m² clear) is a **family WC**: a wall-hung WC in the west cell on the service wall y 9.2 (straight into the shaft), hidden behind a **screen wall** at x 1.5 (y 8.5–9.2) so it is not seen from the gallery when the door (x 2.75, y 8.2–9.1, hinged north, opening in) stands open; a wall-hung basin with its mirror on the screen's east face, seen from the door; the wall on y 7.9 (B12) still closes the stair opening's south side; the gallery and the guard round the opening are untouched.

Bedroom 2 · 4.25 × 3.4 m + its own bath 5.85 m² (3 Nov 2026)

- **Bed 1.4 × 2.0** with its head on the south wall and bedside tables on both sides; **wardrobe 2.0 m** on the east wall (the former niche is the bath's vanity niche); **desk** under the west window. Door from the gallery (swing clear); the bed is in view from it (a 4.25 × 3.4 room: unavoidable, noted).
- **En-suite bath** 4.25 × 1.8 m on axes (≈ 4.7 m² clear with the vanity niche), its door at x 1.7–2.6 on y 11.0 from **inside** the bedroom (hinged east, opening into the bath): a **level walk-in shower 1.0 × 1.6** at the east end with a **linear channel along the service wall** (its trap in the wall), the **wall-hung WC** on the same service wall (y 9.2, shared with the family WC: one frame wall, both outlets west into the shaft), the **vanity** in the west niche under the shaft's south wall; ≈ 2.7 m from the bedroom door. **Result:** every bedroom but one has its own bathroom (the suite, bedrooms 2 and 3) plus a family WC upstairs; bedroom 1 waits for its en-suite (B4 failed, page 3 of its study).

Bedroom 3 · 16.87 m² + its own bath 3.74 m²

- An entry with a wardrobe, then the room: **bed 1.4 × 2.0** head on the west wall with two bedside tables, wardrobe on the east wall, desk on the south wall. Door from the gallery (swing clear).
- **Own bath (en-suite)** off its entry: shower, WC, basin; ≈ 2.7 m from the bedroom door. It stands **exactly over the GF family WC** (walls on the GF lines y 14.4 and 16.1).

Gym route and the owner's office → the L study

- The gallery ends at y 19.25; from there the **gym route** and the **owner's office** belong to the L study (studies/ff_suite.typ), with the gym, the court and the suite.

Interfaces with the L study (fixed)

- ① suite door at the end of the family link (y 6.5); ② bedroom 1 / suite: one straight wall on y 6.5; ③ the gallery's south end (y 19.25), where the gym route starts; ④ bedroom 3's south wall (y 19.25), a wall to the owner's office (no window).

Family part · routes, walls, drains, open items

Routes (≥ 1.0 m clear)

Route	Axes	≈ clear	Deducted
Gallery north (beside the stair)	1.5	1.3	½ wall each side
Gallery on the projection	1.55	1.42	½ wall (rooms side) + ½ window frame W1 on the B3 upstand (void side)
Walk behind the sofa (in the living)	1.1	1	½ wall on the kids' room side; the sofa side is open
Walk beside the sofa to the family link	1.1	1	½ wall on the bedroom 1 side; the sofa side is open
Family link	1.2	1	½ wall each side (bedroom 1 to the north and east)

Walking distance	m
Stair top → kids' room door	2
Stair top → family living (opening)	2.9
Stair top → bedroom 1 door	10.1
Stair top → ⓘ suite door	10.1
Stair top → family WC door	7
Stair top → bedroom 2 door	11.2
Stair top → bedroom 3 door	14.6
Stair top → gallery's south end (L)	18.2
Stair top → lift door	3.3
Bedroom 2 door → its own bath door	2.7
Family living → family WC door	6.4
Bedroom 3 door → its own bath door	2.7
Bedroom 1 door → family WC door	14.9

Stair top → every room, the suite door and the gallery's south end runs only on the arrival strip, the gallery, the family link and the walking line through the living (checked: every door's approach is on circulation, no furniture in a route).

Engineer

- **Projection:** 1.55 m slab beyond x 4.25 along y 6.5–19.25; an **edge beam** on x 5.8 (spans ≈ 6.5 m and 6.25 m, depth to confirm; seen from the courtyard and the living, expressed as the colonnade's beam); 3 **columns** down to their own foundations (sizes and pads to confirm).
- **Ground floor** changes only by the columns, their foundations, the beam line and the lower ceiling under the projection.

Walls off the ground-floor lines

Ground-floor lines here: x 0, 1.25 (guest WC), 1.5 (shaft), 2.25, 2.75, 4.25, 8.95 · y 2.2, 7.9, 10.4, 12.9, 14.4, 16.1, 17.5, 19.25, 21.4 (foyer) (west band); the women's majlis (x 4.25–8.95, y 0–6.5) and the men's majlis (x 2.75–13.311, y 19.25–24) have no inner wall; east part (bedroom 1, family link): over the GF maid wing, laundry and the kitchen's north wall.

- **Off:** y 3.0 (kids' room / living, over the women's majlis); y 11.0 (bedroom 2's bath / bedroom 2); **y 9.2 east of the shaft** (the 0.20 service wall between the family WC and bedroom 2's bath, 3 Nov 2026: light, on the solid west slab); x 2.2 (bedroom 3 bath / entry).
- **On:** y 14.4 and 16.1 (bedroom 3 bath = the GF family WC walls), x 1.5 / y 10.4 (shaft), y 6.5 (bedroom 1 / suite), **x 4.25** (the rooms' east wall, on the GF courtyard edge line), the outer walls.
- The GF riwaq wall x 2.75 carries the floor; the rooms span over the riwaq.

Drains (blue dashed)

- **Family WC and bedroom 2's bath (3 Nov 2026):** every fixture on the shared service wall y 9.2 (the shaft's north wall continued): the two WCs and the basin straight into the shaft, the shower channel's trap in the wall and its outlet west in the wall, the vanity into the shaft's south wall: no pipe in the floor; the floor over the GF utility room and hospitality kitchen (service / wet), nothing in their ceilings.
- **Bedroom 3 bath:** exactly over the GF family WC: WC straight into its stack, shower and basin in the floor build-up. Wet room on a wet room.
- **Bedroom 1:** no bath. No east duct (the suite bath drains to a stack outside the east wall, see ff_suite.typ).

Open items (all in docs/OPEN_ITEMS.md)

- **Engineer:** the edge beam and the 3 columns at x 5.8 (sizes and foundations); walls off the GF lines.
- **Gallery window W1** on the B3 upstand and **W2** on the family living's edge (black steel, divided lites, laminated glass below +4.70): the courtyard stays open to the sky, the house is closed round it (V6.4).
- **Built-in cupboards in niches** (the linen cupboard; bedroom 2's vanity niche): joinery boxes scribed to the walls; decide with the carpenter.
- **Courtyard light:** a daylight study of the courtyard and the double-height living.